Model architecture comparison — Semantic Dial Compass project

How the four model families differ, and how those differences showed up in our experiments.

	GPT-2 small	Phi-3-mini-4k	Gemma-2-2B	Llama-3.2-3B
Release year	2019	2024	2024	2024
Parameters	124 M	3.8 B	2.6 B	3.2 B
Tokenizer	BPE (50257)	tiktoken cl100k-ish (32064)	SentencePiece (256000)	tiktoken (128256)
Layers (L)	12	32	26	28
Attention heads / layer	12 (MHA)	32 (MHA)	8 Q / 4 KV (GQA)	24 Q / 8 KV (GQA)
d_{model}	768	3072	2304	3072
d_{head}	64	96	256	128
Positional encoding	Learned abs.	RoPE	RoPE	RoPE
Attention pattern	full causal	full causal	**sliding window** + full, alternating	full causal
Activation / FFN	GELU	SwiGLU	GeGLU	SwiGLU
Norm type	LayerNorm (post)	RMSNorm (pre)	RMSNorm (pre)	RMSNorm (pre)
Tied embeddings	yes	no	yes (logit softcap)	no
Training data era	WebText (2019)	Phi-3 synthetic + filtered web (2024)	Gemma-2 corpus (2024)	Llama-3 corpus (2024)
Instruct tuned	no	**yes**	base	base
Total attention heads	144	1024	208	672
Primary compass head	**L10 H9 SV0**	**L28 H1 SV0**	**L21 H4 SV0**	**L26 H14 SV2**
Scanned planes (top-4 SVs)	864	1344†	240†	4032
Planes passing both tests	16 (1.9%)	14 (1.0%)	3 (1.2%)	9 (0.2%)
Heads that debias WinoGender	1	1	1	1
Calibrated α (gender)	1.89	6.38	55.26	6.08
PPL Δ at best α	< 1.0	+0.1	+2.0 (at α=2)	+0.035

†Scan restricted to a mid-to-late layer window and head subset selected from a coarser first pass; full enumeration is $L \times H \times \binom{4}{2}$.

— the baseline

Why it works well as a canary. Small, fully open, learned absolute position, post-LN. The scan budget (864 planes) is trivial; you can enumerate everything in minutes on CPU.

Impact on results.

- Cleanest α-sweep: monotone linear, zero near α=1.5, PPL barely moves.
- StereoSet *does* transfer — this is the only model where the gender axis propagates to sentence-level SS.
- Smallest d_{model} $\Rightarrow$ smallest residual norm $\Rightarrow$ smallest α (1.89). α is a calibration scalar, not a claim about signal strength.

Caveat. Old training data; gendered-occupation bias is large ($\approx$ 0.60 `stereo_corr`). Easier target than modern models, so it flatters the method.

— the multilingual payoff

Why it's interesting. Instruct-tuned on heavy synthetic data. RoPE + pre-RMSNorm. Deep (32L) with 1024 total heads, so scan budget explodes; we filtered to late layers.

Impact on results.

- Compass axis decodes to **multilingual gender**: *lui*, , *elle, ella*, , , . Architecture has no multilingual-specific mechanism; this is a pure feature-geometry effect.
- Largest *taxonomy* of pronoun compasses (5 passing heads in the dictionary), because many MHA heads specialise in different pronoun features (number, person, demonstrative).
- StereoSet does *not* transfer — instruct tuning shifts sentence-level scoring, and the compass remains position-local.

Caveat. Filtered / instruction-heavy pretraining changes what probing looks like; watch for prompt-style leakage in ActAdd.

— the specialised one

Why it's different. GQA (8 Q, 4 KV), **sliding-window attention alternating with full, logit softcap**, large $d_{\text{head}} = 256$. Much larger residual norms than the others.

Impact on results.

- **α is huge** (55–81 across domains) — a direct consequence of residual-norm scale, not a weaker compass.
- Head-domain routing is *specialised* (spec index = 1.00): one head (L21 H4) carries most of gender. Good for surgical edits.
- **Religion fails** — 0 of 4 top heads pass the SNR gate; fallback retained. Consistent with the specialisation profile: if the concept isn't localised, routing can't invent a head.
- Sliding-window attention may be why only 3 planes pass out of 240 scanned: short-range context dilutes the pronoun signal.

— the compositional one

Why it stresses the method. GQA (24 Q, 8 KV) means 672 query heads share only 8 KV groups; features smear across many query heads writing into fewer KV slots. Heavy-tailed W_{OV} spectra.

Impact on results.

- Compass rate is $\sim 5\times$ rarer (0.22%) than the others — the compositional / less axis-aligned regime.
- **Gender axis sits at SV2, not SV0.** SV0/SV1 of L26 H14 decode to register / punctuation noise. Same causal sweep, different ordering inside the head. $c_s = 0.53$ cumulative variance — heavier tail than GPT-2 / Phi-3 / Gemma (0.23–0.36).
- **Cleanest intervention by PPL**: +0.035 at α=1.5. Surgical edits survive GQA smearing because the axis is defined in residual-stream coordinates, not head-output coordinates.
- Per-domain routing is *mixed* (spec index $\approx$ 0.47): many heads share each domain $\Rightarrow$ per-domain α and per-domain head selection both help.

How each architectural difference propagated into the experiment pipeline

Architectural choice	Mechanical consequence	Visible in our pipeline as	Mitigation we used
MHA vs GQA (GPT-2 / Phi-3 are MHA; Gemma / Llama are GQA)	GQA: many query heads sharing K/V means features are distributed across query heads; fewer "pure" axis-aligned heads.	Llama's 0.2% pass rate (vs 1–2% for MHA models); gender axis at SV2 for Llama.	Extend scan to top-8 SVs; decode every SV axis, not just SV0 (this is what the taxonomy table shows).
Positional encoding (learned-abs vs RoPE)	RoPE mixes position into Q/K multiplicatively at every attention op; position is not in residual.	Compass axis decoded from W_U gives pronoun tokens only — no position leakage. Phase stability test (scan criterion iii) is unaffected.	None needed; compass is a residual-stream direction, RoPE is orthogonal.
Sliding-window attention (Gemma only)	Alternating local/global layers mean the pronoun signal must re-enter a global layer to reach the compass head.	Gemma's scan planes are few (240) and the passing head is in a global layer (L21). Religion fails — the concept is not localised at a single global layer.	Keep the routed-ensemble top-$K = 4$ heads per domain; if none pass, retain fallback (observed for Gemma religion).
d_{model} **scale** ($768 \to 3072$)	Residual norm grows roughly linearly in d_{model}. SNR $= \|\alpha\sigma u\|/\|r\|$ also scales.	Calibrated α varies from 1.89 (GPT-2) to 55.26 (Gemma). The *relative* edit is similar.	`calibrate_per_domain_alpha.py` solves α so mean SNR hits target budget; keeps cross-model comparisons fair.
Instruct vs base (Phi-3 is instruct; others are base)	Instruct tuning re-shapes sentence-level log-prob distribution; token-level logits at pronoun positions are preserved.	WinoGender (token-level) transfers; StereoSet (sentence-level) does not for Phi-3 / Gemma / Llama.	Report both; treat StereoSet non-transfer as an honest boundary, not a bug.
W_{OV} **spectrum** (GPT-2 light tail; Llama heavy tail)	Heavy-tailed head spectrum ⇒ top-SV directions capture register, not semantics.	Llama gender lives on SV2 of L26 H14; $c_8 = 0.53$ vs ≤ 0.36 for the other models.	Decode 4–8 SVs per head; select the axis by decoded-token semantics, not by rank.
Tokenizer vocabulary (50k BPE vs 128k–256k SentencePiece)	Pronouns (`"he"`, `"she"`) may appear once or many times (capitalised / leading-space variants).	Affects $\ell = W_U u$ decoding — we aggregate across casing variants. Phi-3 / Gemma also show cross-lingual pronoun tokens on the same axis.	Token normalisation in the dictionary reader; cross-lingual evidence treated as a *feature*, not a confound.

What replicates across all four

- **Existence**: exactly one head per model carries a clean gender direction that debias WinoGender.
- **Causality**: monotone linear drop in bias as α increases; sign flip at α=2.0.
- **Surgicality**: PPL cost is small (under 1 point in all four; +0.035 on Llama).
- **Targeted beats diffuse**: compass Pareto-dominates ActAdd in every model at 4–20× lower PPL cost.
- **Dissociation**: other passing compass heads encode number / person / demonstratives, not gender — cherry-pick defence survives.

What varies — and why the architecture explains it

- **Compass rarity**: MHA models pass 1–2% of planes; GQA Llama passes 0.2%. GQA smears features across query heads.
- **SV location**: light-tail spectra (GPT-2, Phi-3, Gemma) put gender on SV0; heavy-tail (Llama) hides it under register noise on SV0/SV1, gender lives at SV2.
- **Domain specialisation**: Gemma (sliding window + GQA) is *specialised*; Phi-3 / Llama are *mixed*. Phi-3's many MHA heads ⇒ richer pronoun-feature taxonomy.
- **StereoSet transfer**: only GPT-2 replicates at sentence-level. Instruct tuning + larger d_{model} decouple token-level and sentence-level bias.
- **Religion failure**: only in Gemma — the specialisation profile has no single head localising religion; routing cannot invent one.